

Substrate Moisture Sensor Volumetric Water Content User Calibration

Index

Substrate Moisture Sensor Volumetric Water Content User Calibration	1
1 Customer Support	3
2 Calibration Procedure	4
2.1 Applicable Device	4
2.2 Calibration Theory	4
2.3 Save calibration data	5
2.3.1 Sensor with RS485 Interface	5
2.3.2 Sensor with SDI-12 Interface	8
2.4 Basic Calibration	10
2.4.1 Fundamentals	11
2.4.2 Save calibration data to sensor with RS485 interface	13
2.4.3 Save calibration data to sensor with SDI-12 interface	20
2.5 Subsection Calibration	20
2.5.1 Fundamentals	20

1 Customer Support

Thank you very much for your order. Our success comes from the continuous faith in the excellence of our products and services, something we are committed to and would never sacrifice. Our customer service, especially in the after sales phase, guarantees the satisfaction of our clients. In line with this strategy, we appreciate that you can share with us your feedback at any time for our improvement, be it positive or negative, so if we can serve you better in anyway, please do inform us.

2 Calibration Procedure

2.1 Applicable Device

The procedure can be applied to substrate moisture sensor SlabSense with RS485 or SDI12 interface.

Here are some abbr.s used in this document.

Abbr.	Meaning	Comment
EP	Epsilon value	Ranges from 1(air) to 81(pure water). For different moisture in substrate, there's different EP value. this value can be read from sensor, please refer to user manual.
VWC	Volumetric Water Content	Ranges from 0.00 to 1.00. Water content in substrate. VWC= Volumetric of water / (Volumetric of sample including air, water and substrate media)

2.2 Calibration Theory

The sensor support user defined curve (UDFC) with your own calibration formula, you can perform a calibration to achieve the best accuracy for specific substrate.

The calibration should be done based on at least 5 substrate samples which with different substrate moisture(VWC) value.

After the sample substrate (e.g. 5 SAMPLEs A/B/C/D/E) are prepared, Insert the sensor into different substrate samples and read the substrate epsilon value (e.g. 5 epsilon values EPa~EPe); then weight the substrate sample and record the value, and dry the substrate samples in oven and then weight again, to get the real VWC value (e.g. 5 VWC values VWCa~VWCe), as shown in following table. You can also use other methods to get the real VWC value.

Sample	Epsilon, Ranges from 1(air) to 81(pure water).	VWC, Ranges from 0.00 to 1.00.
A	EPa	VWCa
B	EPb	VWCb
C	EPc	VWCc
D	EPd	VWCd
E	EPe	VWCe

According to the table above, you can get your own calibration curve, $VWCx=f(EPx)$, in

which f is a function. Using any mathematic tool software or excel to get the polynomial fitting coefficients.

2.3 Save calibration data

The calibration data are all saved in sensor's non-volatile register memory.

Please make sure that the "SUBSTRATE_TYPE" should be set to "2:User Defined Curve", then the substrate volumetric water content calculation will use your calibration formula.

For sensor with RS485 interface, you can write the formula to modbus register by a modbus utility software or SensorOneSet Utility Software, These two utility software will be introduced below.

For sensor with SDI12 interface, you can write the formula using SDI-12 commands, this will be introduced below.

2.3.1 Sensor with RS485 Interface

Parameter	Register Address (HEX)	Data Type	Modbus Function	Parameter Range And comment	Default Value
SUBSTRATE_EPSILON	0x0004 /4	INT16 RO	3/4	0-8188 for 0.00-81.88	N/A
SUBSTRATE_TYPE	0x0020 /32	UINT16 RW	3/6/16	0: Substrate Media 1: Substrateless Media 2: User Defined Curve	0
EPDIV0	0x0060	UINT16 RW	3/4	0-8200 for 0.00~82.00	88
EPDIV1	0x0061	UINT16 RW	3/4	0-8200 for 0.00~82.00	3000
EPDIV2	0x0062	UINT16 RW	3/4	0-8200 for 0.00~82.00	5000
EPDIV3	0x0063	UINT16 RW	3/4	0-8200 for 0.00~82.00	7000

EPDIV4	0x0064	UINT16	3/4	0-8200 for	8188
		RW		0.00~82.00	
A0_EPDIV01	0x0070 0x0071	FLOAT RW	3/4/16	Float value range	-0.053
A1_EPDIV01	0x0072 0x0073	FLOAT RW	3/4/16	Float value range	0.0292
A2_EPDIV01	0x0074 0x0075	FLOAT RW	3/4/16	Float value range	-0.00055
A3_EPDIV01	0x0076 0x0077	FLOAT RW	3/4/16	Float value range	4.3e-006
A0_EPDIV12	0x0078 0x0079	FLOAT RW	3/4/16	Float value range	-0.053
A1_EPDIV12	0x007A 0x007B	FLOAT RW	3/4/16	Float value range	0.0292
A2_EPDIV12	0x007C 0x007D	FLOAT RW	3/4/16	Float value range	-0.00055
A3_EPDIV12	0x007E 0x007F	FLOAT RW	3/4/16	Float value range	4.3e-006
A0_EPDIV23	0x0080 0x0081	FLOAT RW	3/4/16	Float value range	-0.053
A1_EPDIV23	0x0082 0x0083	FLOAT RW	3/4/16	Float value range	0.0292
A2_EPDIV23	0x0084 0x0085	FLOAT RW	3/4/16	Float value range	-0.00055
A3_EPDIV23	0x0086 0x0087	FLOAT RW	3/4/16	Float value range	4.3e-006
A0_EPDIV34	0x0088 0x0089	FLOAT RW	3/4/16	Float value range	-0.053
A1_EPDIV34	0x008A 0x008B	FLOAT RW	3/4/16	Float value range	0.0292
A2_EPDIV34	0x008C 0x008D	FLOAT RW	3/4/16	Float value range	-0.00055
A3_EPDIV34	0x008E 0x008F	FLOAT RW	3/4/16	Float value range	4.3e-006



UINT16:16 bit unsigned integer

INT16:16bit signed integer

RO: Register is Read Only

R/W: Register is Read/Write

HEX: Hexadecimal (data with 0x/0X prefix)

DEC: Decimal

FLOAT: If IEEE754 little-endian byte order is [A,B,C,D], the FLOAT format is [B,A,D,C]

FLOAT_INVERSE: If IEEE754 little-endian byte order is [A,B,C,D], the FLOAT_INVERSE format is [D,C,B,A]

2.3.2 Sensor with SDI-12 Interface

Request	Response	Comment
aR1!, aRC1!	Response data format : a<±SubstrateTemp><+SubstrateVWC><+SubstrateECBulk><+SubstrateCalibratedRawCounts><+SubstrateVWCEpsilon><+SubstrateECPore>[<CRC>]<CR><LF>	Example: Request: 0R1! Response: 0+24.1+40.50+1620+2888.77+25.47+5972<CR><LF>
aXR_SUBSTRATETYPE!	aSUBSTRATETYPE=<X> <X> is substrate type: 0: Substrate 1: Substrateless 2: UDF(User Defined Curve) The substrate type is only used for <SubstrateVWC> calculation based on different formula.	Query substrate type Example: Request: 0XR_SUBSTRATETYPE! Response: 0SUBSTRATETYPE=2<CR><LF>
aXW_SUBSTRATETYPE_<X>!	aSUBSTRATETYPE=<X>	Configure substrate type Example: Request: 0XW_SUBSTRATETYPE_2! Response: 0SUBSTRATETYPE=0<CR><LF>
aXR_UDFC_D!	aUDFC_D=< EPDIV0>,< EPDIV1>,< EPDIV2>,< EPDIV3>,< EPDIV4>! <EPDIV> value range is 0-8200 for 0.00~82.00	Query User Defined Curve Epsilon Division point Example: Request : 0XR_UDFC_D! Respons : 0UDFC_D=+88,+3000,+5000,+7000,+8188
aXW_UDFC_D_<EPDIV0>,<EPDIV1>,<EPDIV2>,<EPDIV3>,<EPDIV4>!	aUDFC_D=< EPDIV0>,< EPDIV1>,< EPDIV2>,< EPDIV3>,< EPDIV4>! < EPDIV> value range is 0-8200 for 0.00~82.00	Configure User Defined Curve Epsilon Division point, using command below to write 0.88, 30.00, 50.00, 70.00, 81.88 to sensor. Example: Request : 0XW_UDFC_D_88,3000,5000,7000,8188!

		Respons : 0UDFC_D=+88,+3000,+5000,+7000,+8188
aXR_UDFC_0!	aUDFC_0=<A0_EPDIV01>,<A1_EP DIV01>,<A2_EP DIV01>,<A3_EPDI V01>! <A0_EP DIV01>,<A1_EP DIV01>,<A 2_EP DIV01>,<A3_EP DIV01> are the polynomial coefficients for epsilon range from EPDIV0 to EPDIV1.	Query User Defined Curve polynomial coefficients for epsilon range from EPDIV0 to EPDIV1. Example: Request : 0XR_UDFC_0! Respons : 0UDFC_0=-5.2999999E-02,+2.9200001E-02,-5.50000 00E-04,+4.3000000E-06
aXW_UDFC_0 _<A0_EP DIV0 1>,<A1_EP DI V01>,<A2_EP DIV01>,<A3_E PDIV01>!	aUDFC_0=<A0_EP DIV01>,<A1_EP DIV01>,<A2_EP DIV01>,<A3_EPDI V01>! <A0_EP DIV01>,<A1_EP DIV01>,<A 2_EP DIV01>,<A3_EP DIV01> are the polynomial coefficients for epsilon range from EPDIV0 to EPDIV1.	Configure User Defined Curve polynomial coefficients for epsilon range from EPDIV0 to EPDIV1. Example: Request : 0XW_UDFC_0 -5.3E-02,+2.92E-02,-5.5E-04,+4.3E-0 6! Response : 0UDFC_0=-5.2999999E-02,+2.9200001E-02,-5.50000 00E-04,+4.3000000E-06
aXR_UDFC_1!	aUDFC_1=<A0_EP DIV12>,<A1_EP DIV12>,<A2_EP DIV12>,<A3_EPDI V12>! <A0_EP DIV12>,<A1_EP DIV12>,<A 2_EP DIV12>,<A3_EP DIV12> are the polynomial coefficients for epsilon range from EPDIV1 to EPDIV2.	Query User Defined Curve polynomial coefficients for epsilon range from EPDIV1 to EPDIV2. Example: Request : 0XR_UDFC_1! Respons : 0UDFC_1=-5.2999999E-02,+2.9200001E-02,-5.50000 00E-04,+4.3000000E-06
aXW_UDFC_1 _<A0_EP DIV1 2>,<A1_EP DI V12>,<A2_EP DIV12>,<A3_E PDIV12>!	aUDFC_1=<A0_EP DIV12>,<A1_EP DIV12>,<A2_EP DIV12>,<A3_EPDI V12>! <A0_EP DIV12>,<A1_EP DIV12>,<A 2_EP DIV12>,<A3_EP DIV12> are the polynomial coefficients for epsilon range from EPDIV1 to EPDIV2.	Configure User Defined Curve polynomial coefficients for epsilon range from EPDIV1 to EPDIV2. Example: Request : 0XW_UDFC_1 -5.3E-02,+2.92E-02,-5.5E-04,+4.3E-0 6! Response : 0UDFC_1=-5.2999999E-02,+2.9200001E-02,-5.50000 00E-04,+4.3000000E-06

aXR_UDFC_2!	aUDFC_2=<A0_EPDIV23>,<A1_EP DIV23>,<A2_EPDIV23>,<A3_EPDI V23>!	Query User Defined Curve polynomial coefficients for epsilon range from EPDIV2 to EPDIV3. Example: Request : 0XR_UDFC_2!
	<A0_EPDIV23>,<A1_EPDIV23>,<A 2_EPDIV23>,<A3_EPDIV23> are the polynomial coefficients for epsilon range from EPDIV2 to EPDIV3.	Respons : 0UDFC_2=-5.2999999E-02,+2.9200001E-02,-5.50000 00E-04,+4.3000000E-06
aXW_UDFC_2 _<A0_EPDIV2 3>,<A1_EPDI V23>,<A2_EP DIV23>,<A3_E PDIV23>!	aUDFC_2=<A0_EPDIV23>,<A1_EP DIV23>,<A2_EPDIV23>,<A3_EPDI V23>! <A0_EPDIV23>,<A1_EPDIV23>,<A 2_EPDIV23>,<A3_EPDIV23> are the polynomial coefficients for epsilon range from EPDIV2 to EPDIV3.	Configure User Defined Curve polynomial coefficients for epsilon range from EPDIV2 to EPDIV3. Example: Request : 0XW_UDFC_2_-5.3E-02,+2.92E-02,-5.5E-04,+4.3E-0 6! Response : 0UDFC_2=-5.2999999E-02,+2.9200001E-02,-5.50000 00E-04,+4.3000000E-06
aXR_UDFC_3!	aUDFC_3=<A0_EPDIV34>,<A1_EP DIV34>,<A2_EPDIV34>,<A3_EPDI V34>! <A0_EPDIV34>,<A1_EPDIV34>,<A 2_EPDIV34>,<A3_EPDIV34> are the polynomial coefficients for epsilon range from EPDIV3 to EPDIV4.	Query User Defined Curve polynomial coefficients for epsilon range from EPDIV3 to EPDIV4. Example: Request : 0XR_UDFC_3! Respons : 0UDFC_3=-5.2999999E-02,+2.9200001E-02,-5.50000 00E-04,+4.3000000E-06
aXW_UDFC_3 _<A0_EPDIV3 4>,<A1_EPDI V34>,<A2_EP DIV34>,<A3_E PDIV34>!	aUDFC_3=<A0_EPDIV34>,<A1_EP DIV34>,<A2_EPDIV34>,<A3_EPDI V34>! <A0_EPDIV34>,<A1_EPDIV34>,<A 2_EPDIV34>,<A3_EPDIV34> are the polynomial coefficients for epsilon range from EPDIV3 to EPDIV4.	Configure User Defined Curve polynomial coefficients for epsilon range from EPDIV3 to EPDIV4. Example: Request : 0XW_UDFC_3_-5.3E-02,+2.92E-02,-5.5E-04,+4.3E-0 6! Response : 0UDFC_3=-5.2999999E-02,+2.9200001E-02,-5.50000 00E-04,+4.3000000E-06

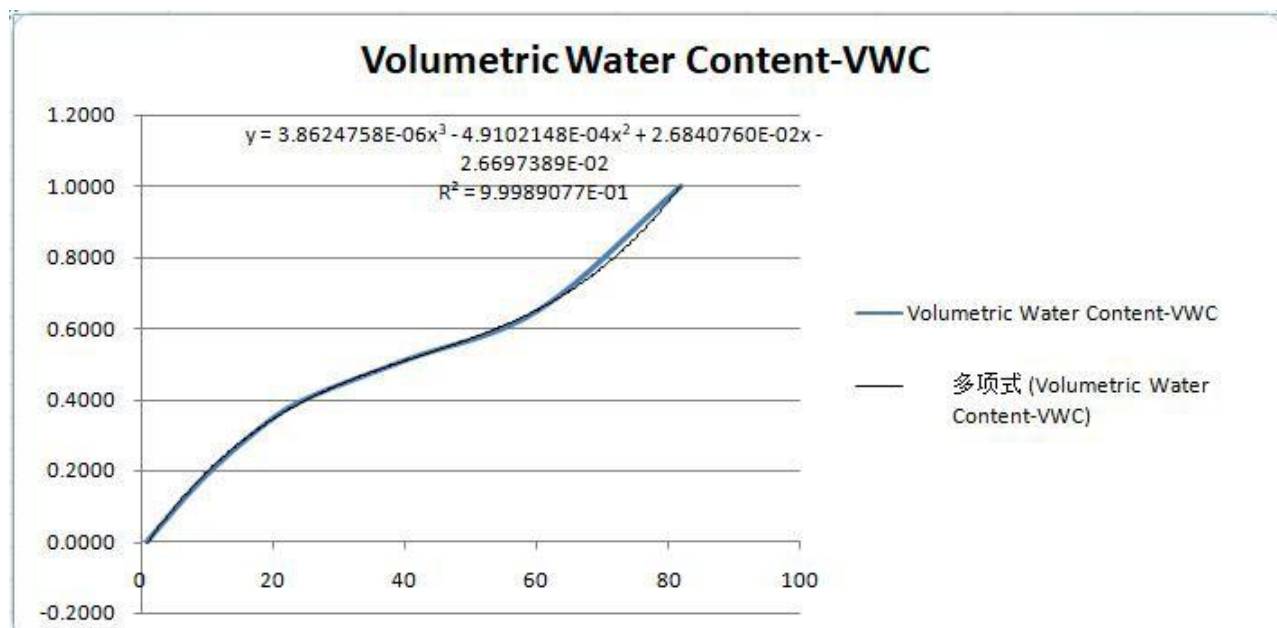
2.4 Basic Calibration

2.4.1 Fundamentals

Suppose we have prepared 5 substrate samples with different substrate moisture as listed in table below, and insert the sensor into each substrate sample and then read 5 E_p value from the sensor, NOTE this VWC is not the value read from sensor but being calculated from each substrate sample by oven drying and weighing method.

Sample	Ep-Epsilon	VWC-Moisture	Comment
A	0.88	0%(0.00)	Sensor in Air
B	9.10	17.35%(0.1735)	Sensor in sample 1
C	18.16	32.16%(0.3216)	Sensor in sample 2
D	25.21	40.30%(0.4030)	Sensor in sample 3
E	40.15	51.10%(0.5110)	Sensor in sample 4
F	60.31	65.11%(0.6511)	Sensor in sample 5
G	81.88	100%(1.00)	Sensor in pure water

Now we use EXCEL to get a curve and then using polynomial fitting to get coefficients.



Step 3: Summarize all coefficients to write to sensor				
Epsilon Division Point	Epsilon Value			
EPDIV0	0.88			
EPDIV1	18.16			
EPDIV2	40.15			
EPDIV3	60.31			
EPDIV4	81.88			
Epsilon Segment Range	Coff A0	Coff A1	Coff A2	Coff A3
EPDIV0-EPDIV1	A0_EPDIV01	A1_EPDIV01	A2_EPDIV01	A3_EPDIV01
0.88-18.16	-2.6697389E-02	2.6840760E-02	-4.9102148E-04	3.8624758E-06
EPDIV1-EPDIV2	A0_EPDIV12	A1_EPDIV12	A2_EPDIV12	A3_EPDIV12
18.16-40.15	-2.6697389E-02	2.6840760E-02	-4.9102148E-04	3.8624758E-06
EPDIV2-EPDIV3	A0_EPDIV23	A1_EPDIV23	A2_EPDIV23	A3_EPDIV23
40.15-60.31	-2.6697389E-02	2.6840760E-02	-4.9102148E-04	3.8624758E-06
EPDIV3-EPDIV4	A0_EPDIV34	A1_EPDIV34	A2_EPDIV34	A3_EPDIV34
60.31-81.88	-2.6697389E-02	2.6840760E-02	-4.9102148E-04	3.8624758E-06

You can get:

$$y = 3.8624758E-06x^3 - 4.9102148E-04x^2 + 2.6840760E-02x - 2.6697389E-02$$

That is:

$$VWC = 3.8624758E-06*EP^3 - 4.9102148E-04*EP^2 + 2.6840760E-02*EP - 2.6697389E-02$$

A common formula is:

$$VWC = A0 + A1*Epsilon^1 + A2*Epsilon^2 + A3*Epsilon^3$$

Then coefficients are:

$$A0 = -2.6697389E-02; A1 = +2.6840760E-02; A2 = -4.9102148E-04; A3 = 3.8624758E-06$$

Then we make a table as below:

Epsilon Division Point	Epsilon Value
EPDIV0	0.88
EPDIV1	18.16
EPDIV2	40.15
EPDIV3	60.31
EPDIV4	81.88

Epsilon Segment Range	Coffs
EPDIV0 to 1 : 0.88~18.16	A0EPDIV01= -2.6697389E-02; A1EPDIV01= +2.6840760E-02; A2EPDIV01= -4.9102148E-04; A3EPDIV01= +3.8624758E-06
EPDIV1 to 2 : 18.16~40.15	A0EPDIV12= -2.6697389E-02; A1EPDIV12= +2.6840760E-02; A2EPDIV12= -4.9102148E-04; A3EPDIV12= +3.8624758E-06
EPDIV2 to 3 : 40.15~60.31	A0EPDIV23= -2.6697389E-02; A1EPDIV23= +2.6840760E-02; A2EPDIV23= -4.9102148E-04; A3EPDIV23= +3.8624758E-06
EPDIV3 to 4 : 60.31~81.88	A0EPDIV34= -2.6697389E-02; A1EPDIV34= +2.6840760E-02; A2EPDIV34= -4.9102148E-04; A3EPDIV34= +3.8624758E-06

2.4.2 Save calibration data to sensor with RS485 interface

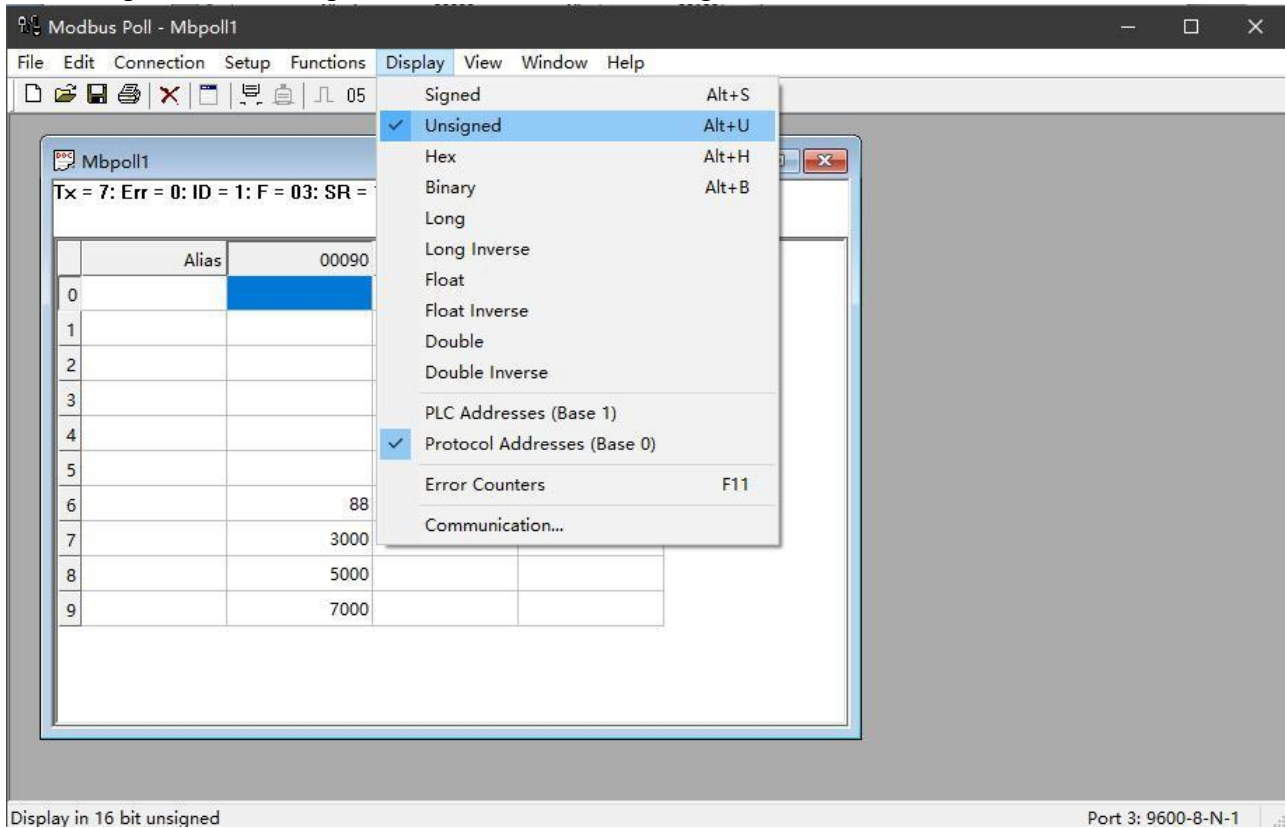
When we get calibration above, we write those value into sensor, by modbus command.

- For Epsilon Division point: 0.88, 18.16, 40.15, 60.31 and 81.88, we should write 88, 1816, 4015, 6031 and 8188 into register 0x0060-0x0064.
- For Coffs of curve EPDIV0 to 1: we should write -2.6697389E-02, +2.6840760E-02, -4.9102148E-04, 3.8624758E-06 into register 0x0070-0x0077.
- For Coffs of curve EPDIV1 to 2: we should write -2.6697389E-02, +2.6840760E-02, -4.9102148E-04, 3.8624758E-06 into register 0x0078-0x007F.
- For Coffs of curve EPDIV2 to 3: we should write -2.6697389E-02, +2.6840760E-02, -4.9102148E-04, 3.8624758E-06 into register 0x0080-0x0087.
- For Coffs of curve EPDIV3 to 4: we should write -2.6697389E-02, +2.6840760E-02, -4.9102148E-04, 3.8624758E-06 into register 0x0088-0x008F.

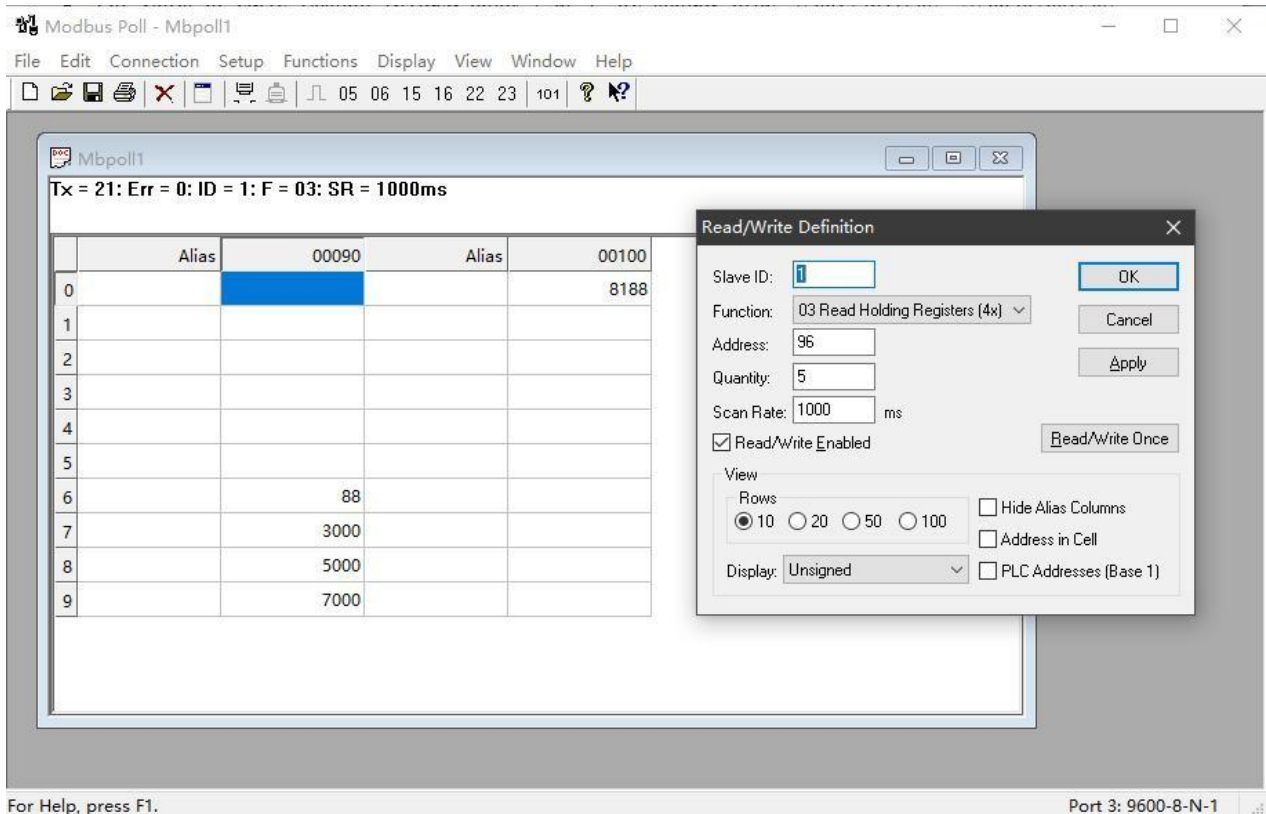
- Note that 0x0060-0x0064 should be read/write together. And 0x0070-0x008F should be read/write together. Those register does not support read/write separately.
- Then you should write to substrate type register (0x0020) with value 2(User Defined Curve) to make the calibration formula effective.

2.4.2.1 Write calibration data to sensor by “Modbus Poll” Utility

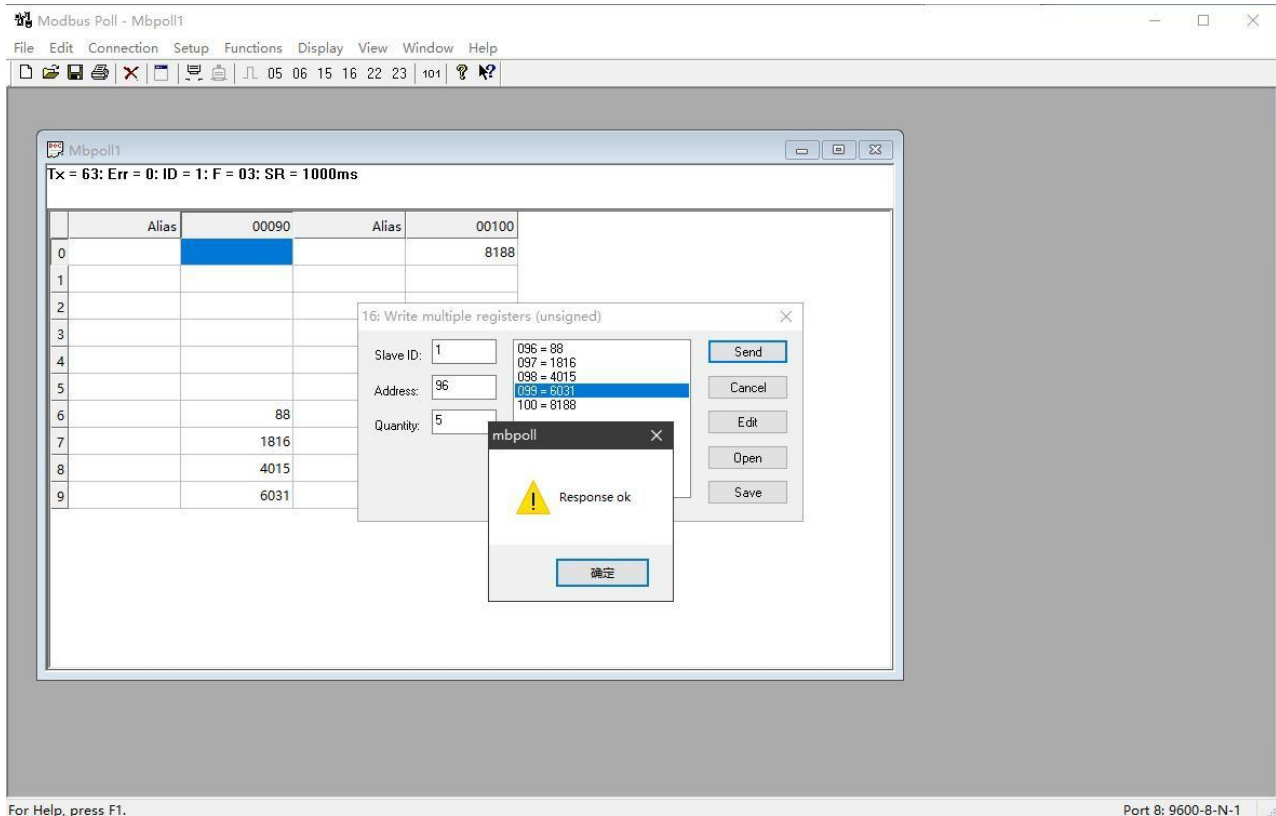
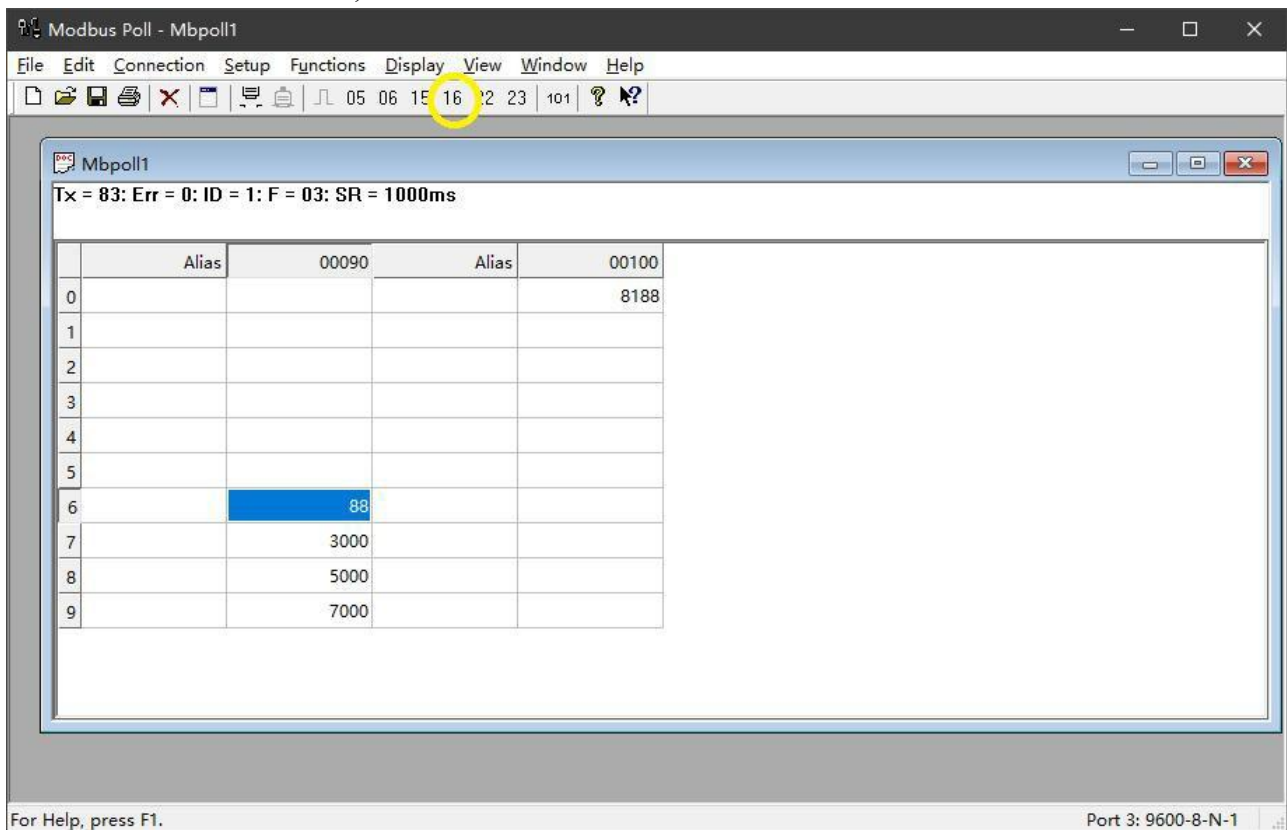
- For Epsilon Division point, Set data format as unsigned



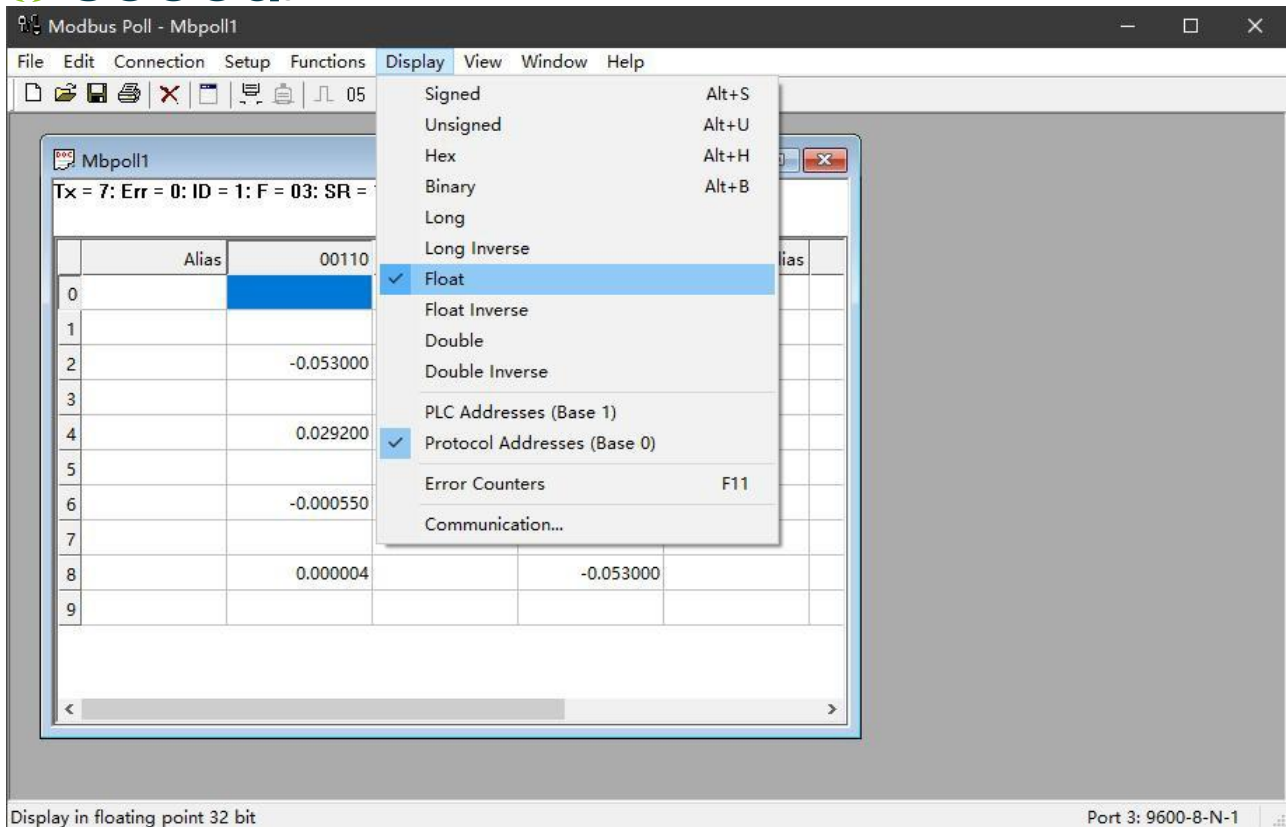
Define the read/write register and quantity



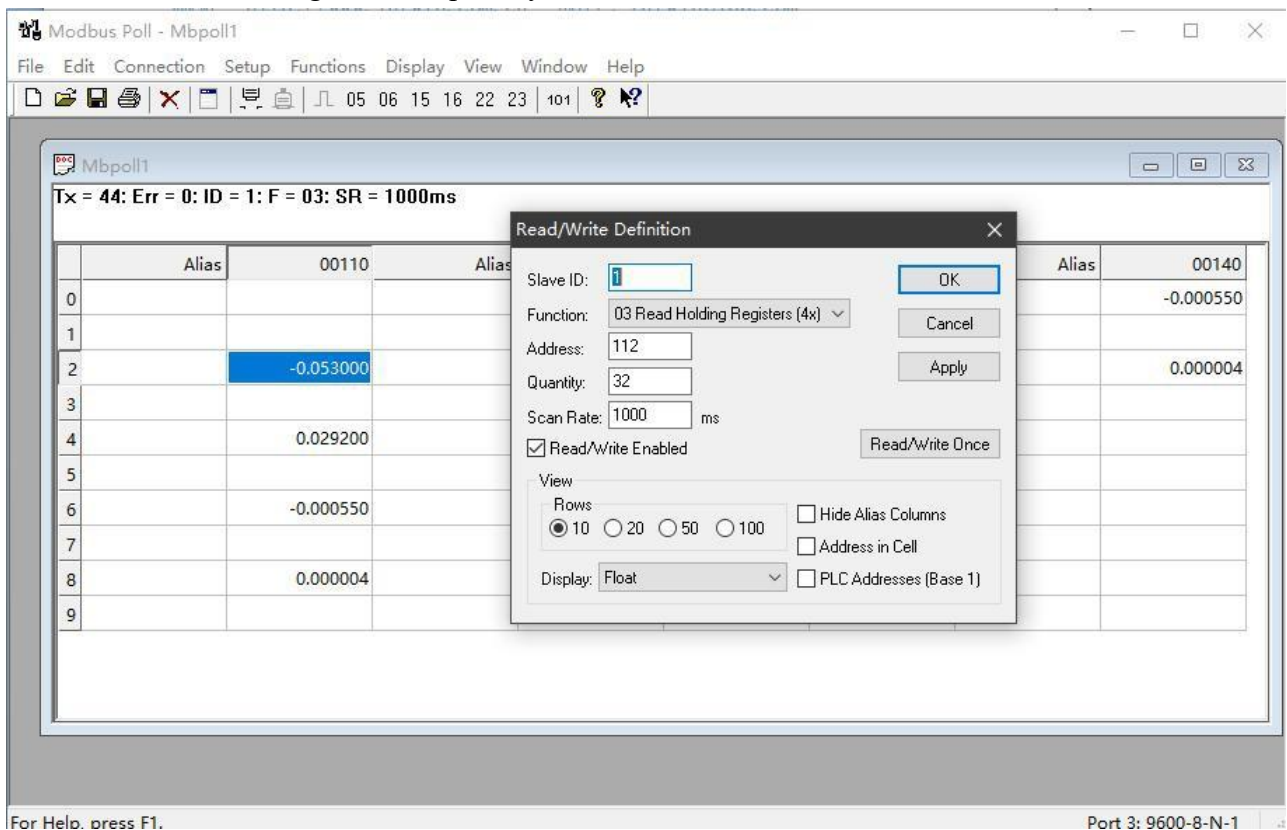
Click on button 16 and edit, then send



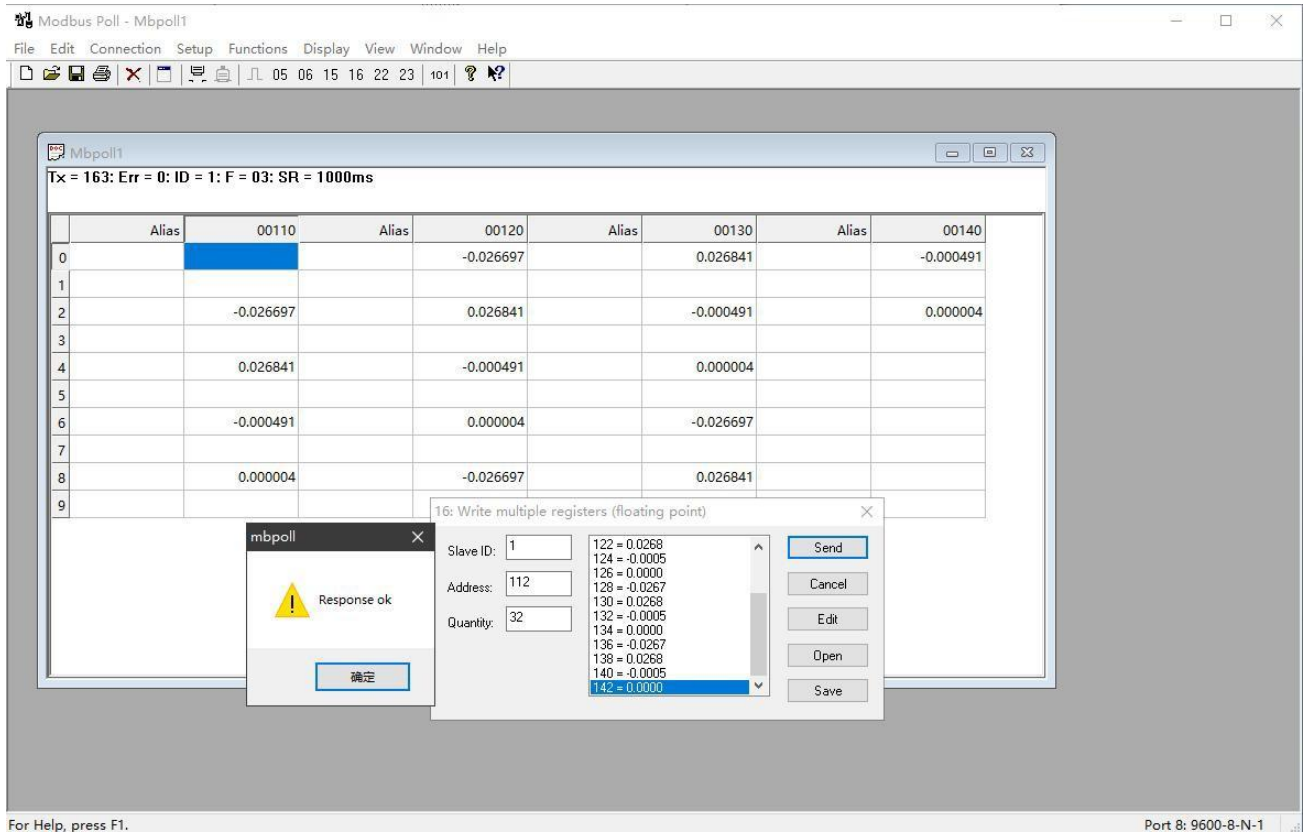
● For Coffs: Set data format as float



Define the read/write register and quantity



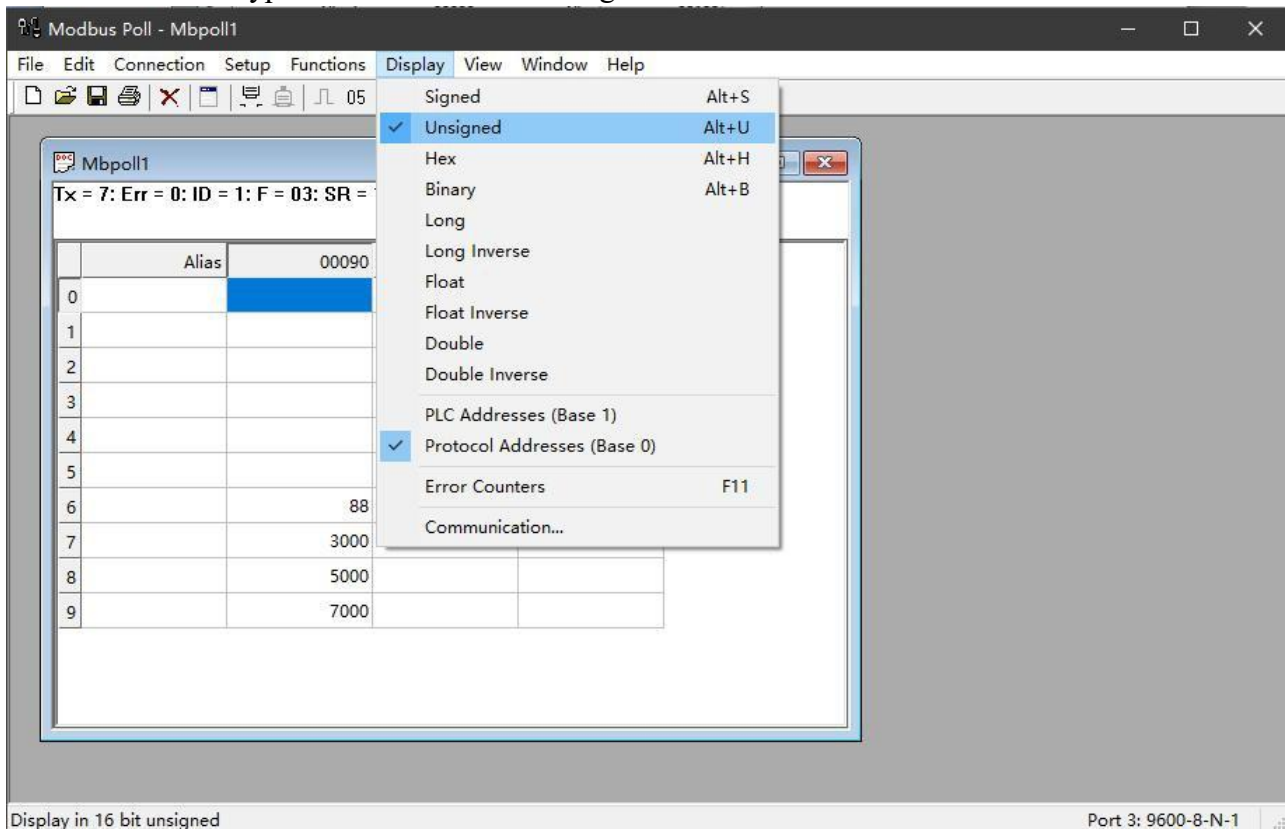
Click on button 16 and edit, then send



For Help, press F1.

Port 8: 9600-8-N-1

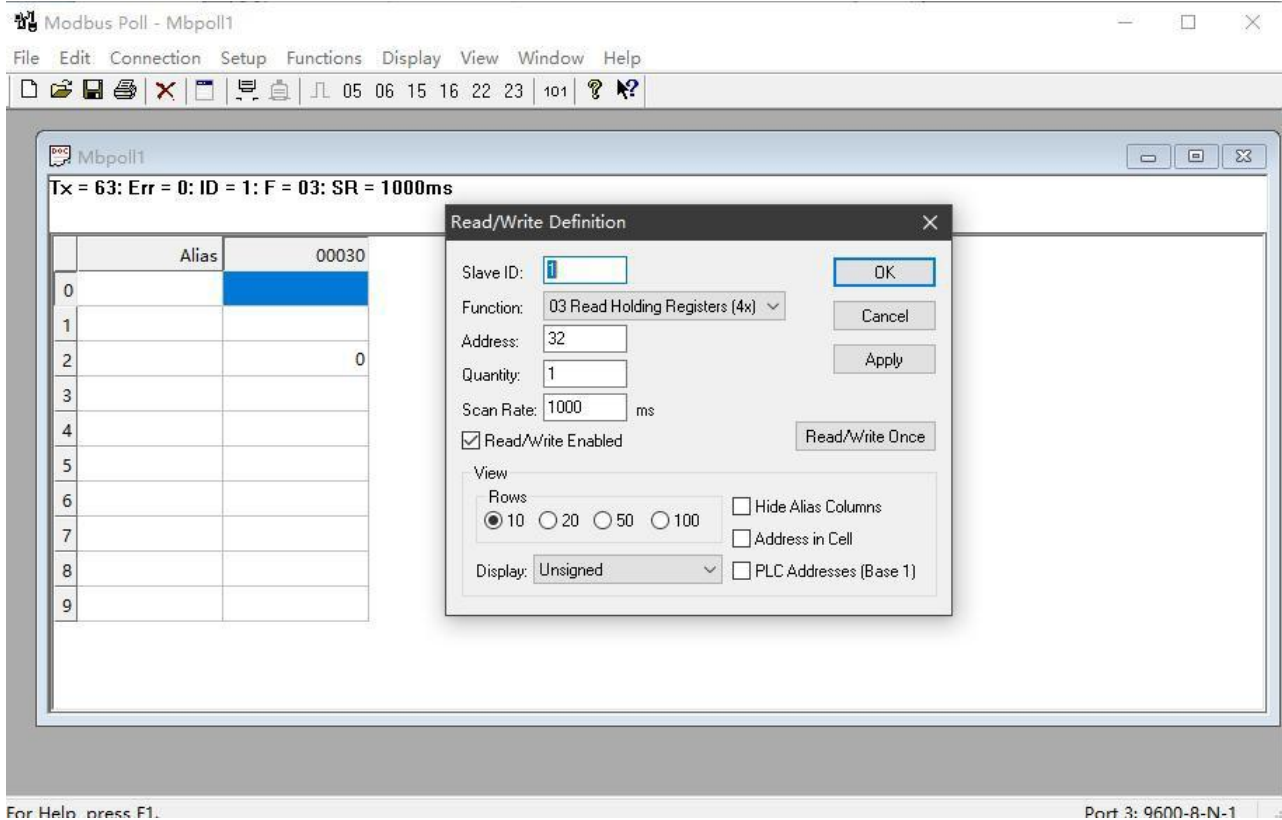
● For Substrate Type: Set data format as unsigned



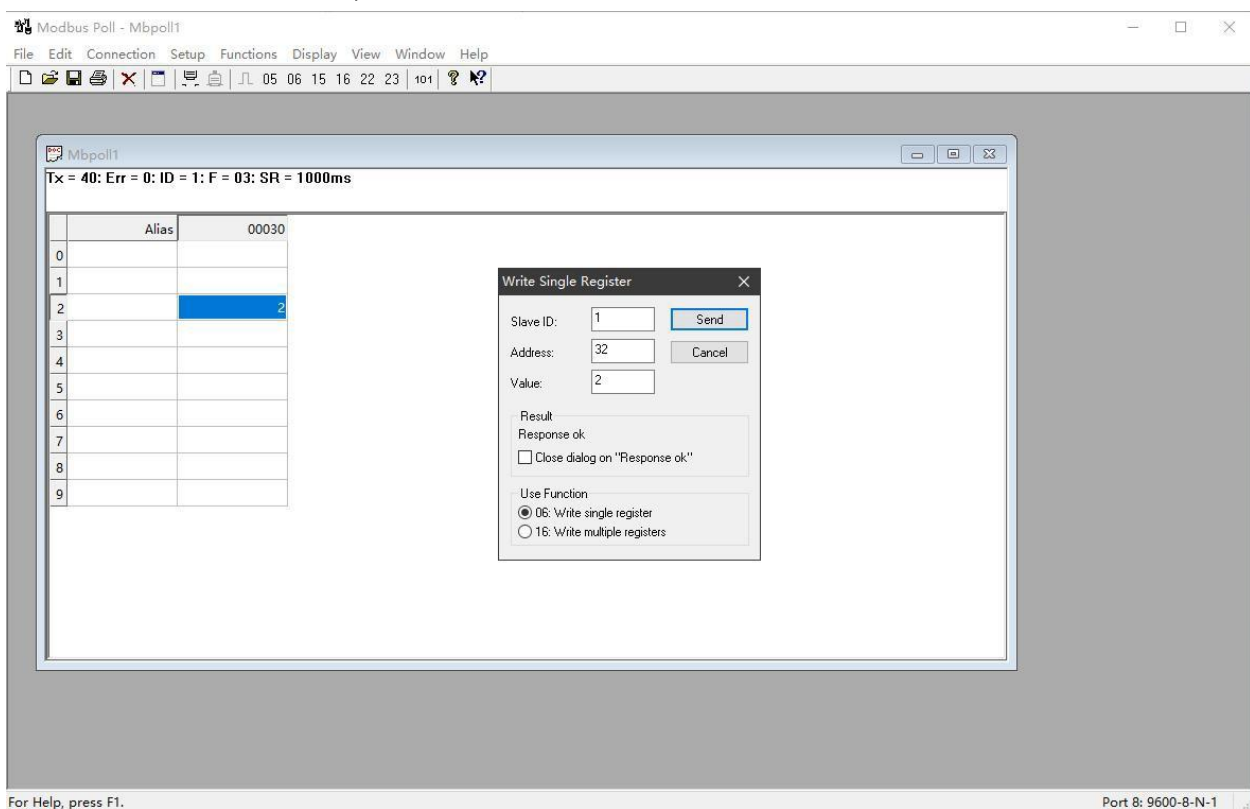
Display in 16 bit unsigned

Port 3: 9600-8-N-1

Define the read/write register and quantity



Click on button 16 and edit, then send



2.4.3 Save calibration data to sensor with SDI-12 interface

Comment
Configure User Defined Curve Epsilon Division point, using command below to write 0.88, 18.16, 40.15, 60.31, 81.88 to sensor. Example: Request : 0XW_UDFC_D_88,1816,4015,6031,8188! Respons : 0UDFC_D=+88,+1816,+4015,+6031,+8188
Configure User Defined Curve polynomial coefficients for epsilon range from EPDIV0 to EPDIV1. Example: Request : 0XW_UDFC_0_-2.6697389E-02,2.6840760E-02,-4.9102148E-04,+3.8624758E-06! Response : 0UDFC_0=-2.6697389E-02,2.6840760E-02,-4.9102148E-04,+3.8624758E-06
Configure User Defined Curve polynomial coefficients for epsilon range from EPDIV1 to EPDIV2. Example: Request : 0XW_UDFC_1_-2.6697389E-02,2.6840760E-02,-4.9102148E-04,+3.8624758E-06! Response : 0UDFC_1=-2.6697389E-02,2.6840760E-02,-4.9102148E-04,+3.8624758E-06
Configure User Defined Curve polynomial coefficients for epsilon range from EPDIV2 to EPDIV3. Example: Request : 0XW_UDFC_2_-2.6697389E-02,2.6840760E-02,-4.9102148E-04,+3.8624758E-06! Response : 0UDFC_2=-2.6697389E-02,2.6840760E-02,-4.9102148E-04,+3.8624758E-06
Configure User Defined Curve polynomial coefficients for epsilon range from EPDIV3 to EPDIV4. Example: Request : 0XW_UDFC_3_-2.6697389E-02,2.6840760E-02,-4.9102148E-04,+3.8624758E-06! Response : 0UDFC_3=-2.6697389E-02,2.6840760E-02,-4.9102148E-04,+3.8624758E-06
Configure substrate type Request: 0XW_SUBSTRATETYPE_2! Response: 0SUBSTRATETYPE=0<CR><LF>

2.5 Subsection Calibration

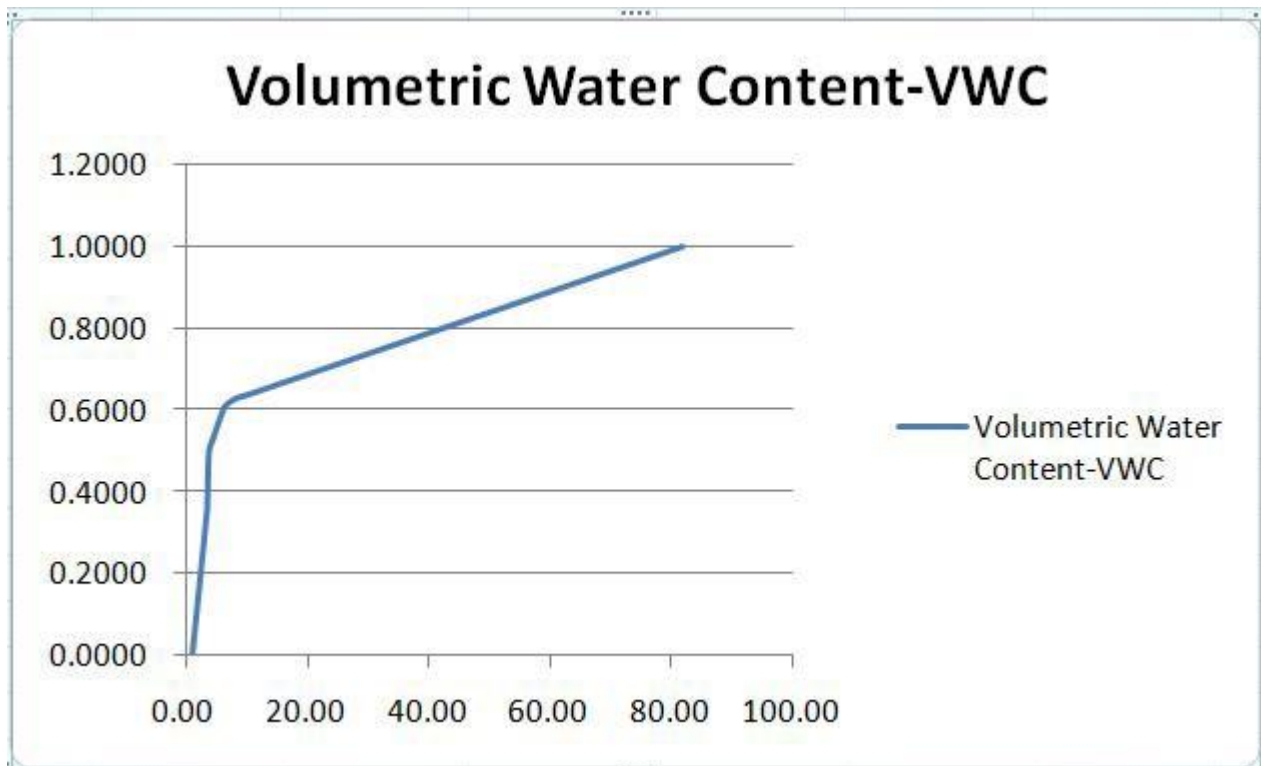
2.5.1 Fundamentals

For some kind of growing media, a single formula cannot fit the data for best result, then you should split the curve and make several formula to get several group of coefficients.

Sample	Ep-Epsilon	WVC-Moisture	Comment
A	0.88	0%(0.0000)	Sensor in Air
B	3.13	32.72% (0.3272)	Sensor in sample 1
C	3.38	37.22% (0.3722)	Sensor in sample 2
D	3.61	49.57% (0.4957)	Sensor in sample 3

E	4.22	52.65% (0.5265)	Sensor in sample 4
F	5.66	58.82% (0.5882)	Sensor in sample 5
G	6.40	61.11% (0.6111)	Sensor in sample 6
H	8.38	63.00% (0.6300)	Sensor in sample 7
I	9.65	64.00% (0.6400)	Sensor in sample 8
J	81.88	100% (1.0000)	Sensor in pure water

And the curve as below, it's hard to make single formula for best fit the curve, so you should split the curve into several curves, and do polynomial fitting for each curve.

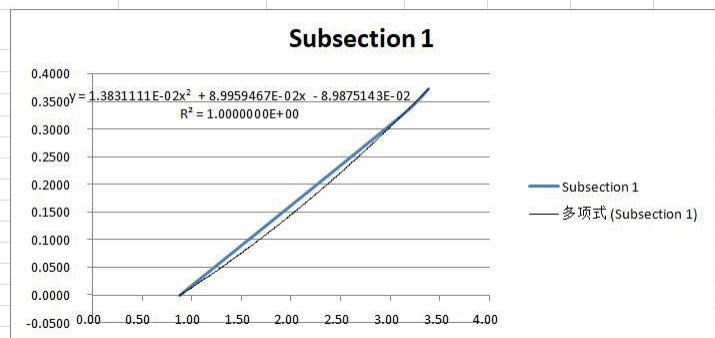


● We split the curve above into 4 curves as below

Subsection 1

Step 2: Polynomial fit (Subsection 1)

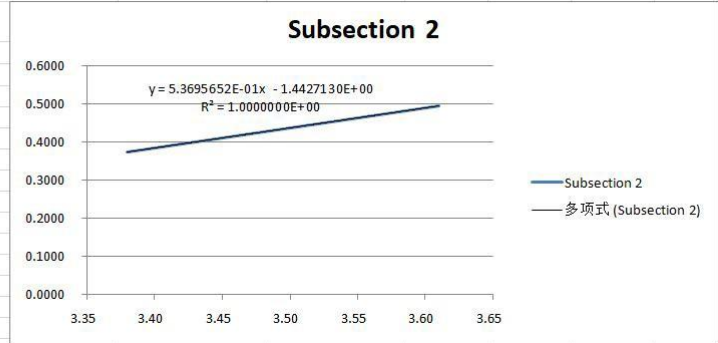
Epsilon-EP	Volumetric Water Content-VWC
0.88	0.0000
3.13	0.3272
3.38	0.3722



Subsection 2

Step 2: Polynomial fit (Subsection 2)

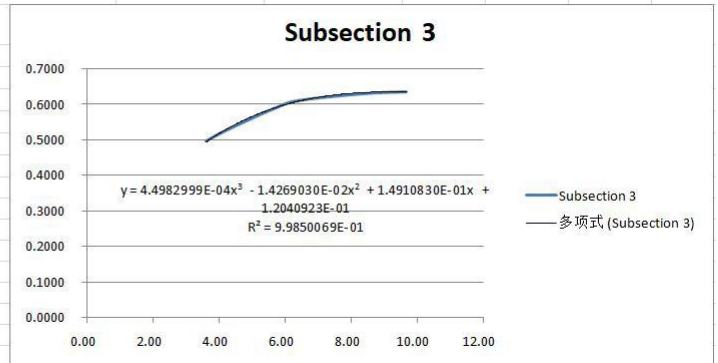
Epsilon-EP	Volumetric Water Content-VWC
3.38	0.3722
3.61	0.4957



Subsection 3

Step 2: Polynomial fit (Subsection 3)

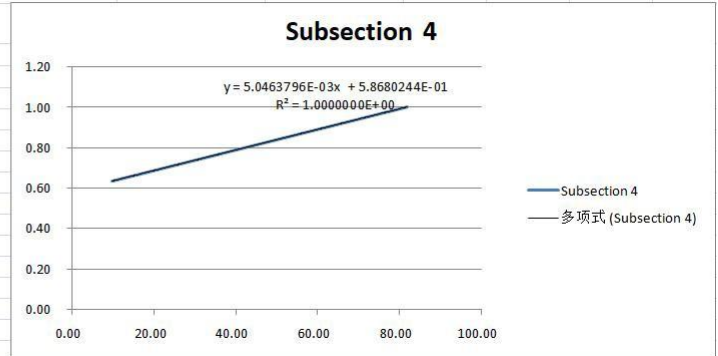
Epsilon-EP	Volumetric Water Content-VWC
3.61	0.4957
4.22	0.5265
5.66	0.5882
6.40	0.6111
8.38	0.63
9.65	0.64



Subsection 4

Step 2: Polynomial fit (Subsection 4)

Epsilon-EP	Volumetric Water Content-VWC
9.65	0.64
81.88	1.00



- Then we can get several groups of coefficients for each curves as listed in table below, you can write those data to sensor as described in basic calibration.

Step 3: Summarize all coefficients to write to sensor

Epsilon Division Point	Epsilon Value			
EPDIV0	0.88			
EPDIV1	3.38			
EPDIV2	3.61			
EPDIV3	9.65			
EPDIV4	81.88			
Epsilon Segment Range	Coff A0	Coff A1	Coff A2	Coff A3
EPDIV0-EPDIV1	A0_EPDIV01	A1_EPDIV01	A2_EPDIV01	A3_EPDIV01
0.88-3.38	-8.9875143E-02	8.9959467E-02	1.3831111E-02	0.0000000E+00
EPDIV1-EPDIV2	A0_EPDIV12	A1_EPDIV12	A2_EPDIV12	A3_EPDIV12
3.38-3.61	-1.4427130E+00	5.3695652E-01	0.0000000E+00	0.0000000E+00
EPDIV2-EPDIV3	A0_EPDIV23	A1_EPDIV23	A2_EPDIV23	A3_EPDIV23
3.61-9.65	1.2040923E-01	1.4910830E-01	-1.4269030E-02	4.4982999E-04
EPDIV3-EPDIV4	A0_EPDIV34	A1_EPDIV34	A2_EPDIV34	A3_EPDIV34
9.65-81.88	5.8680244E-01	5.0463796E-03	0.0000000E+00	0.0000000E+00

Then we make a table as below:

Epsilon Division Point	Epsilon Value
EPDIV0	0.88
EPDIV1	3.38
EPDIV2	3.61
EPDIV3	9.65
EPDIV4	81.88

Epsilon Segment Range	Coffs
EPDIV0 to 1 : 0.88-3.38	A0EPDIV01=-8.9875143E-02; A1EPDIV01=8.9959467E-02; A2EPDIV01=1.3831111E-02; A3EPDIV01=0.0000000E+00;
EPDIV1 to 2 : 3.38-3.61	A0EPDIV12=-1.4427130E+00; A1EPDIV12=5.3695652E-01; A2EPDIV12=0.0000000E+00; A3EPDIV12=0.0000000E+00;
EPDIV2 to 3 : 3.61-9.65	A0EPDIV23=1.2040923E-01; A1EPDIV23=1.4910830E-01; A2EPDIV23=-1.4269030E-02; A3EPDIV23=4.4982999E-04;
EPDIV3 to 4 : 9.65-81.88	A0EPDIV34=5.8680244E-01; A1EPDIV34=5.0463796E-03; A2EPDIV34=0.0000000E+00; A3EPDIV34=0.0000000E+00;